

LORENZO BINDI

ROBOTICS SOFTWARE ENGINEER

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Date of birth: 9 December 2001



PROFILE

Robotics Engineer with an MSc in Electrical and Automation Engineering and hands-on experience in robot learning, dexterous manipulation, teleoperation and feedback control. Currently an Electrical Engineer Consultant at ALTEN Italy for Leonardo. Combines software development for physical robots with practical experience in electronic systems, PCB debugging, testing and engineering documentation.

PROFESSIONAL EXPERIENCE

ALTEN Italy | Jan 2026 – Present

Electrical Engineer Consultant

Client: **Leonardo** | Montevarchi, Italy

- Support electronic systems, working with power distribution systems and electrical/electronic system documentation.
- Read and analyse PCB schematics to support debugging, validation and troubleshooting; perform debugging and testing on electronic boards.
- Use Xpedition and PLM tools for electronic design review and engineering workflow support.

EDUCATION

University of Florence, Italy | Oct 2023 – Dec 2025

Master's Degree in Electrical and Automation Engineering

Automation and Robotics curriculum | **110/110 cum laude**

Thesis: Development and Implementation of Learning Techniques Based on Human Demonstrations for the Shadow Hand Robot.

University of Florence, Italy | 2020 – 2023 | **104/110**

Bachelor's Degree in Electronics and Telecommunications Engineering

Thesis: Convolutional Neural Networks for Image Analysis.

CNN design and training, image classification, preprocessing and augmentation.

TECHNICAL SKILLS

Programming and systems: Python, C/C++, MATLAB; Linux, ROS, UDP communication.

Robot learning: Imitation learning, Behavior Cloning, recurrent policies (GRU); experimental reinforcement learning (PPO, SAC).

Control and robotics: Kinematics, dynamics, PID, MPC, CEM optimisation, teleoperation, joint command mapping and filtering.

Perception: OpenCV, ArUco pose estimation, quaternion processing, HSV colour segmentation, sensor integration.

Simulation: MuJoCo, Gymnasium Robotics, Gazebo, RViz, Simulink, Simscape Multibody.

ML frameworks: PyTorch, TensorFlow, Keras, NumPy, Scikit-learn.

Electronics and EDA: Power distribution systems, PCB schematic analysis, board debugging and testing; Xpedition, PLM.

Relevant Coursework: Robotics; Control Systems; Estimation and Identification; Machine Learning; Reinforcement Learning; Computer Vision; Embedded Systems; Multi-Agent Systems.

Languages: Italian — Native; English — B2 (Cambridge FCE); French — B1.

Certifications: Cambridge First Certificate in English (B2), 2019.

Interests: Motocross, football.

SELECTED PROJECTS

Dexterous Manipulation from Human Demonstrations

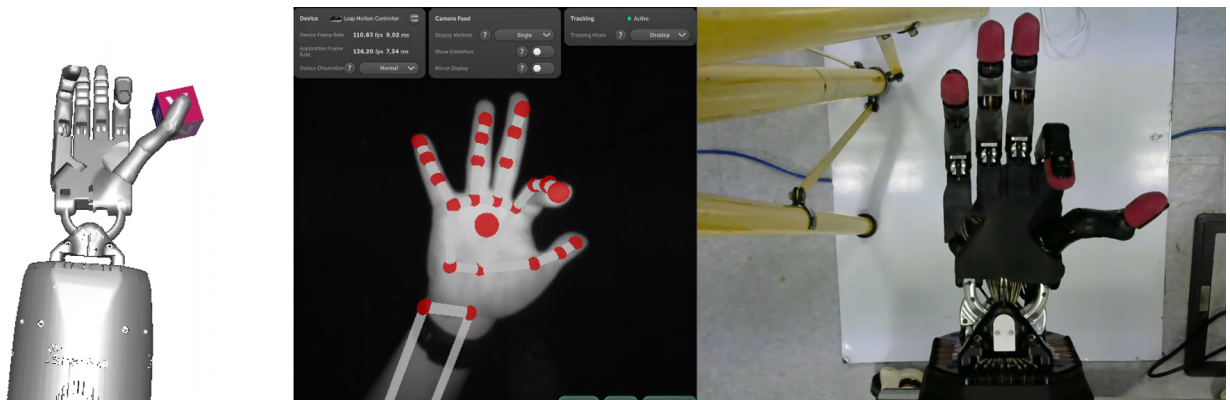
Master's thesis | University of Florence | 2025

Python; Leap Motion; Shadow Hand; MuJoCo; Gymnasium Robotics; GRU / Behavior Cloning; OpenCV

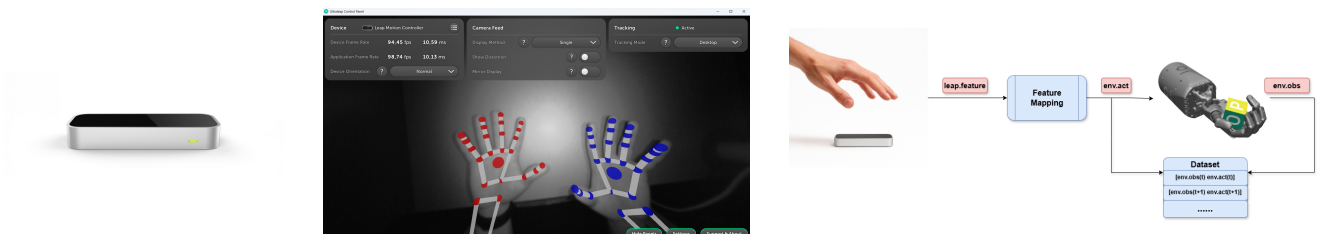
Developed a pipeline to teach a high-DOF robotic hand to grasp, hold and rotate a cube from human demonstrations, then combined the learned behaviour with control strategies for simulation and physical execution.

- **Teleoperation and robot interface.** Extracted finger and wrist features from Leap Motion, calibrated the human-to-robot mapping and generated 22 joint commands. Implemented normalised-to-radian conversion, joint limits, exponential smoothing and slew-rate limiting, with UDP command transmission at approximately 100 Hz on the physical system.
- **Demonstration dataset and learning.** Recorded state–action sequences and trained a GRU-based Behavior Cloning policy using recent observation history. Applied feature normalisation, Huber loss, validation-based early stopping and data augmentation to learn temporally coherent commands.
- **Perception and state estimation.** Integrated camera-based ArUco pose estimation, quaternion filtering and angular-velocity estimation. Used a compact six-feature state on hardware, including object position, tilt, rotation speed and marker confidence, with handling for temporary occlusions.
- **Hybrid control.** Compared a greedy one-step controller with Behavior Cloning and their combination in simulation. On hardware, implemented predictive control with Cross-Entropy Method (CEM) sampling, joint masks and finger-dependent blending with the learned policy.
- **Experimental evaluation.** Compared simulated behaviour over sets of 5, 10 and 20 episodes and tested the full perception–action loop on the physical hand. Investigated RL-based extensions and characterised limitations from observation noise, latency, grasp stability and contact dynamics.

Observed result: best physical trials achieved approximately two complete cube rotations in just over three minutes while maintaining the grasp. Reliability varied between trials; occlusion, latency and sim-to-real mismatch remained key constraints.



Three connected views: simulated Shadow Hand (left), Leap Motion hand tracking (centre), and the physical Shadow Hand (right).



Leap Motion Controller (left), tracked hand skeletons (centre), and demonstration-data pipeline (right).

ADDITIONAL PROJECTS

Collaborative Robot Programming – Franka Challenge

University of Florence | Team project | 2024

Python; ROS; Franka Emika Panda; OpenCV; ArUco; HSV; Gazebo / RViz

- Developed reusable Python ROS nodes and keyboard teleoperation for a 7-DOF Panda robot, publishing Cartesian linear and angular velocity commands.
- Performed pick-and-place of four cubes and tuned gripper parameters for the manipulation task.
- Processed end-effector camera images with OpenCV to detect ArUco markers using the $6 \times 6_{1000}$ dictionary and identify cube colours through tuned HSV thresholds.

Result: demonstrated keyboard-driven manipulation, marker recognition and colour segmentation on the physical setup.

Quadrotor Modelling, Attitude Control and Flight Tests

University of Florence | Team project | 2024

MATLAB; Simulink; Betaflight; Cascaded PID; IMU data analysis

- Modelled nonlinear 6-DOF dynamics, motor response, thrust/torque mixing, actuator saturation and sensor noise in Simulink.
- Designed and tuned cascaded attitude and angular-rate PID loops, mapped the gains to Betaflight and compared simulated responses with physical flight-test logs.
- Analysed time-stamped IMU and radio-control data, converting quaternions to Euler angles to assess reference tracking.
- Investigated drift in the logging IMU and improved consistency with the flight-controller IMU by adjusting complementary-filter weighting.

Result: obtained flight responses comparable to the default tuning and identified a sensor-estimation issue through targeted bench tests.

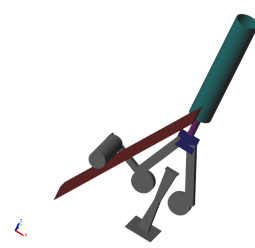
Hydraulic Gripper – Multibody and Contact Simulation

University of Florence | Academic team project

MATLAB / Simulink; Simscape Multibody; Hydraulic actuation; Contact modelling

- Built a parameterised 3D model of a hydraulically actuated gripper for grasping a test tube, coupling a cylinder–piston mechanism to the linkage.
- Modelled sliding and gripper–object contacts, hydraulic actuation and a feedback control loop.
- Studied friction, contact stiffness and solver step size; refined an initial multiple-contact formulation using a simplified contact model and constrained object motion.

Result: demonstrated the grasping mechanism in simulation and examined the trade-off between contact detail and numerical stability.



Left: Panda robot and cubes. Centre: physical quadrotor. Right: simulated hydraulic gripper.